

# Wireless Communication Exam Notes

EX 715 (BEI) / EX 751 (BEX) · IV-I · TU, IOE · tiers over 22 papers, 2070–2082 BS

## 2. Radio Wave Propagation in Mobile Network Environment

12 hrs, **the largest chapter in the syllabus** · asked in 22 of 22 papers · 7 syllabus sub-topics · **by far the heaviest numerical load** → Ch 3 numerical companion.

**Read this first:**

- The chapter splits cleanly in two, and the exam treats them separately.
  - **Large-scale path loss:** how far the signal reaches, averaged over hundreds of wavelengths. Free space, two-ray, Okumura, Hata, indoor.
  - **Small-scale fading:** how the signal flutters over a few wavelengths. Delay spread, Doppler, coherence bandwidth, Rayleigh / Rician.
- Three questions are near-certain: **small-scale fading types + factors** **TOP 6/22**, **coherence bandwidth and coherence time** **HOT 5/22**, and an **Okumura or Hata numerical** **HOT 4/22**.
- Two derivations are worth memorising line by line, because they recur: the **two-ray path difference and phase difference** **HOT 4/22** and the **Doppler shift** **HOT 3/22**.
- **Nearly every question here has a numerical twin.** Learn the theory with the companion open.

### 3.1 Free space propagation

**Explain free space propagation. Relate radiated power to electric field** **8** **2+2+2+2** **6**

**HOT 3/22** 80 Ba, 81 Ba, 78 Ch, 80 Bh · **almost always set as a numerical** → Ch 3 companion

**When it applies**

- Used to predict received signal strength when Tx and Rx have a **clear, unobstructed LOS path**.
- Eg satellite links, microwave LOS links.
- **Almost never true for a mobile in a city**, but every other model is built as a correction to it.

**Step 1: power flux density**

- An **isotropic** source radiates  $P_t$  uniformly over a sphere of area  $4\pi d^2$ .
- **Power flux density** at distance  $d$ :

$$P_d = \frac{P_t G_t}{4\pi d^2} \quad \text{W/m}^2$$

- **EIRP** =  $P_t G_t$  watts: the power an isotropic antenna would need to give the same reading in the main beam.
- **ERP** uses a half-wave dipole as the reference instead:  $\text{ERP} = \text{EIRP} - 2.15 \text{ dB}$ .

**Step 2: effective aperture**

- **Effective aperture**  $A_e$  = ratio of power available at the antenna terminals to the incident power per unit area.

$$A_e = \frac{G_r \lambda^2}{4\pi}, \quad \lambda = \frac{c}{f}$$

**Step 3: the Friis equation**

- $P_r = P_d \cdot A_e$ , so

$$P_r(d) = \frac{P_t G_t G_r \lambda^2}{(4\pi)^2 d^2 L}$$

- $L \geq 1$  = system loss (feeder attenuation, filters, mismatch).  $L = 1$  means no hardware loss.
- $P_r \propto 1/d^2$  **and**  $\propto \lambda^2$ : doubling distance costs 6 dB, doubling frequency costs 6 dB.

**Relating power to E-field (the 78 Ch / 80 Ba style numerical)**

- Power flux density in terms of the field:

$$P_d = \frac{|E|^2}{\eta} = \frac{|E|^2}{120\pi}, \quad \eta = 377 \, \Omega \text{ (free space)}$$

- So  $E$  falls as  $1/d$  while power falls as  $1/d^2$ .
- Received power through the aperture:

$$P_r(d) = \frac{|E|^2}{120\pi} \cdot \frac{G_r \lambda^2}{4\pi}$$

- **rms voltage** into a matched receiver of input resistance  $R_{\text{ant}}$ :

$$P_r = \frac{V_{\text{rms}}^2}{R_{\text{ant}}} = \frac{(V_{\text{ant}}/2)^2}{R_{\text{ant}}} \Rightarrow V_{\text{ant}} = 2\sqrt{P_r R_{\text{ant}}}$$

**The factor 2 is the matched-load voltage divider. Losing it is the standard mistake.**

**Path loss**

- **With** antenna gains:

$$PL(\text{dB}) = 10 \log \frac{P_t}{P_r} = -10 \log \left[ \frac{G_t G_r \lambda^2}{(4\pi)^2 d^2} \right]$$

- **Without** antenna gains ( $G_t = G_r = 1$ ):

$$PL(\text{dB}) = -10 \log \left[ \frac{\lambda^2}{(4\pi)^2 d^2} \right]$$

- Useful dB form,  $d$  in km and  $f$  in MHz:

$$PL_{\text{FS}} = 32.44 + 20 \log f_{\text{MHz}} + 20 \log d_{\text{km}}$$

verified/added

- The minus sign on the gain term says that term is a **gain**. The  $(4\pi d/\lambda)^2$  term is the **free space loss** proper.
- Raising  $d$  raises the loss, which pushes designers to **lower frequencies** to keep the link budget manageable.

**Far field (Fraunhofer region)**

- Friis is valid only **beyond** the far-field distance

$$d_f = \frac{2D^2}{\lambda}$$

where  $D$  = the largest physical linear dimension of the antenna.

- Also needs  $d_f \gg D$  and  $d_f \gg \lambda$ .
- The equation blows up at  $d = 0$ , so a **close-in reference distance**  $d_0$  is used:

$$P_r(d) = P_r(d_0) \left( \frac{d_0}{d} \right)^2, \quad d \geq d_0 \geq d_f$$

- $d_0$  is typically **1 km** for large cells, **100 m** for microcells, **1 m** indoors.

### 3.2 The three propagation mechanisms

Explain the three basic radio wave propagation mechanisms. Large scale Vs small scale model 3+6 3

8

HOT 3/22 72 Ash, 73 Bh, 72 Ma

- Apart from the direct wave, the receiver gets **reflected**, **diffracted** and **scattered** waves.
- They add **vectorially**, so the resultant varies in strength in real time. That is the physical origin of fading.

#### 1. Reflection

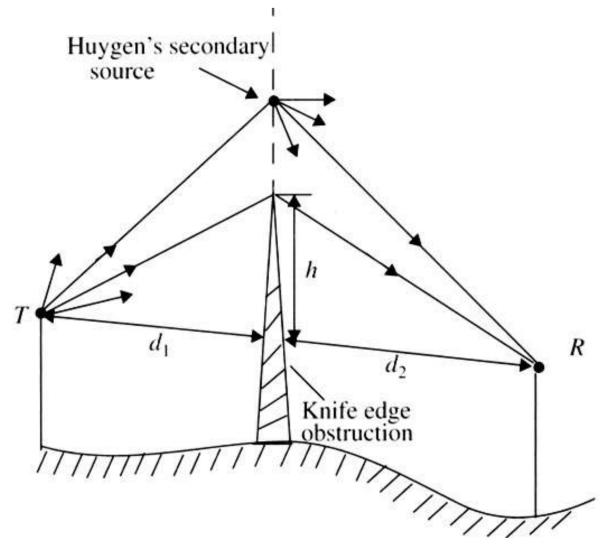
- Wave hits an object **very large compared to the wavelength**.
- Eg the earth's surface, buildings, walls.
- Part reflects, part transmits into the object.
- Strength set by the **reflection coefficient**  $\Gamma$ , which depends on polarisation, angle of incidence and the material.
- At grazing incidence  $\Gamma \rightarrow -1$ , ie **full reflection with a  $180^\circ$  phase flip**. That is the assumption the two-ray model rests on.

#### 2. Diffraction (79 Bh pairs this with a Fresnel derivation, 2+6 )

- Path is obstructed by a surface with **sharp edges**.
- Secondary waves bend around the obstacle into the shadow region.
- **This is what lets a mobile work with no LOS at all** in urban and hilly terrain.
- Explained by **Huygens' principle**: every point on a wavefront acts as a point source of secondary wavelets, which add to form the new front.
- Diffraction loss grows as the obstruction fills more of the first Fresnel zone.

#### 3. Scattering

- Medium contains objects **small compared to the wavelength**, or rough surfaces.
- Eg foliage, street signs, lamp posts, rain drops, snow.
- Energy is **spread in many directions**.
- Follows the same principles as diffraction.
- **Received signal is often stronger than reflection and diffraction alone predict**, because scattering fills in the gaps.



Knife-edge diffraction geometry. Effective obstruction height  $h$ , distances  $d_1$  and  $d_2$  from Tx and Rx.

### Fresnel zones and the diffraction parameter

- A **Fresnel zone** is the ellipsoid of revolution over which secondary waves reach the receiver with an excess path of  $n\lambda/2$ .
- Numbered  $F_1, F_2, F_3, \dots$ . Odd zones add constructively, even zones destructively.
- Size set by **frequency** and **Tx-Rx distance**.
- **Fresnel-Kirchhoff diffraction parameter**

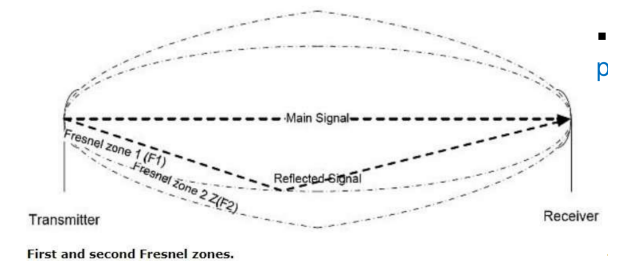
$$v = h \sqrt{\frac{2(d_1 + d_2)}{\lambda d_1 d_2}}$$

- Radius of the  $n$ -th zone at distance  $d_1$ :

$$r_n = \sqrt{\frac{n\lambda d_1 d_2}{d_1 + d_2}}$$

verified/added

- **Rule of thumb: keep 55 % of the first Fresnel zone clear** and the link behaves as free space.
- Diffraction gain  $G_d$  (dB) against  $v$ :
  - $v \leq -1$ :  $G_d = 0$
  - $-1 \leq v \leq 0$ :  $G_d = 20 \log(0.5 - 0.62v)$
  - $0 \leq v \leq 1$ :  $G_d = 20 \log(0.5 e^{-0.95v})$
  - $v > 2.4$ :  $G_d = 20 \log(0.225/v)$



First and second Fresnel zones.

First and second Fresnel zones.

### 3.3 Two-ray ground reflection model

## Derive the path difference and phase difference in the two-ray model, and the path loss 8 2+6 2+8 6

**HOT 4/22** 71 Bh, 74 Bh, 76 Bh, 81 Bh · 76 Bh asks first for the (+) over free space, so open with that

### Why it beats the free space model (the 2-mark opener)

- Free space assumes **one** path. A real mobile link always has a **ground-reflected** path as well.
- The two-ray model adds that second path, so it predicts the  $1/d^4$  **roll-off actually measured** over large distances, instead of  $1/d^2$ .
- Found **reasonably accurate** for large-cell mobile systems over a few km.
- Needs no measured data, unlike Okumura or Hata.

### Assumptions

- $h_t, h_r \gg \lambda$  and  $h_t, h_r \ll d$ .
- Reflected wave arrives at **grazing incidence**, so  $\Gamma = -1$ .
- Flat earth, no curvature.

### Step 1: path difference $\Delta$ (method of images)

- Replace the reflection by an **image transmitter** at  $-h_t$ .
- LOS path and reflected path become two straight lines:

$$d' = \sqrt{(h_t - h_r)^2 + d^2}, \quad d'' = \sqrt{(h_t + h_r)^2 + d^2}$$

- So

$$\Delta = d'' - d' = d\sqrt{1 + \left(\frac{h_t + h_r}{d}\right)^2} - d\sqrt{1 + \left(\frac{h_t - h_r}{d}\right)^2}$$

- Expand each root binomially, valid since  $d \gg h_t, h_r$ :

$$\Delta \approx d \left\{ \left(1 + \frac{1}{2} \left(\frac{h_t + h_r}{d}\right)^2\right) - \left(1 + \frac{1}{2} \left(\frac{h_t - h_r}{d}\right)^2\right) \right\}$$

$$\Delta \approx \frac{2h_t h_r}{d}$$

### Step 2: phase difference and time delay

$$\theta_\Delta = \frac{2\pi\Delta}{\lambda} = \frac{4\pi h_t h_r}{\lambda d}$$

$$\tau_d = \frac{\Delta}{c} = \frac{\theta_\Delta}{2\pi f_c}$$

### Step 3: total E-field

- For large  $d$ ,  $|E_0 d_0 / d'| \approx |E_0 d_0 / d''| \approx |E_0 d_0 / d|$ , so the two rays differ **only in phase**:

$$|E_{\text{tot}}(d)| = \frac{E_0 d_0}{d} |\cos \theta_\Delta - \cos 0|$$

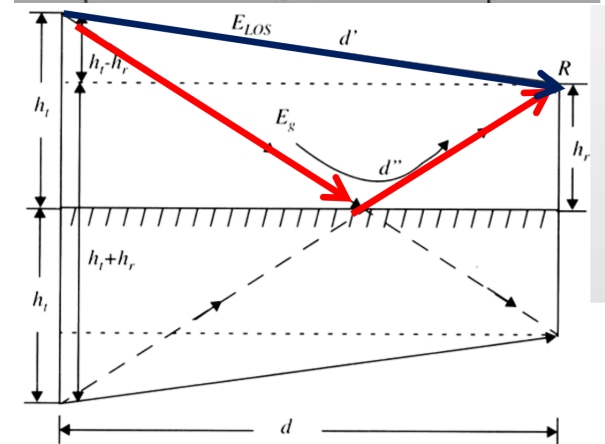
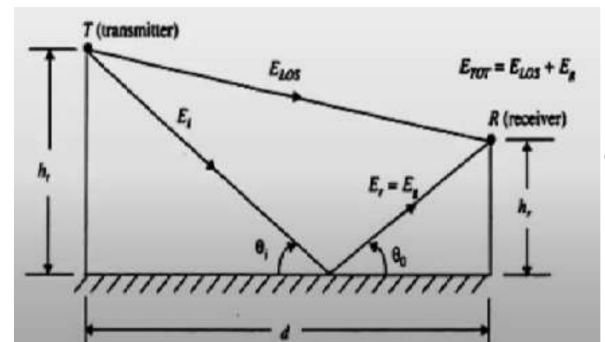
- Using  $\cos \alpha - \cos \beta$  identity:

$$|E_{\text{tot}}(d)| = 2 \frac{E_0 d_0}{d} \sin\left(\frac{\theta_\Delta}{2}\right)$$

- For  $\theta_\Delta/2 < 0.3$  rad,  $\sin x \approx x$ :

$$|E_{\text{tot}}(d)| \approx \frac{2E_0 d_0}{d} \cdot \frac{2\pi h_t h_r}{\lambda d} = \frac{k}{d^2}$$

- The field now falls as  $1/d^2$ , so power falls as  $1/d^4$ .**



Method of images: the ground-reflected ray is redrawn as a straight line from the image transmitter at  $-h_t$ .

### Step 4: received power and path loss

- Using  $P_r = |E_{\text{tot}}|^2 A_e / \eta$ :

$$P_r = P_t G_t G_r \frac{h_t^2 h_r^2}{d^4}$$

- Note  $\lambda$  has cancelled:** the two-ray result is **independent of frequency**.
- In dB:

$$PL(\text{dB}) = 40 \log d - (10 \log G_t + 10 \log G_r + 20 \log h_t + 20 \log h_r)$$

- Valid only beyond the **crossover distance**  
 $d_c = 4\pi h_t h_r / \lambda$ ; below it the model oscillates and free space is used instead. verified/added

### 3.4 Practical link budget design

## Explain the log-distance path loss model and log-normal shadowing 4 2+4

PIN 1/22 77 Ch · the **link budget numericals** built on it are 10 and 12 marks (74 Bh, 76 Bh) → Ch 3 companion

### Log-distance path loss model

- Both theory and measurement show received power falls **logarithmically** with distance, indoors and outdoors.
- Generalises free space by letting the exponent be  $n$  instead of 2:

$$\overline{PL}(d) \propto \left(\frac{d}{d_0}\right)^n$$

$$\overline{PL}(d) [\text{dB}] = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right)$$

- $\overline{PL}(d_0)$  is computed by the **free space** formula, or measured.
- Bars mean **ensemble average** over all possible paths at that distance.
- One number,  $n$ , absorbs the whole environment.**

| Environment                   | Path Loss Exponent, $n$ |
|-------------------------------|-------------------------|
| Free space                    | 2                       |
| Urban area cellular radio     | 2.7 to 3.5              |
| Shadowed urban cellular radio | 3 to 5                  |
| In building line-of-sight     | 1.6 to 1.8              |
| Obstructed in building        | 4 to 6                  |
| Obstructed in factories       | 2 to 3                  |

### Log-normal shadowing

- Problem: two locations at the **same**  $d$  can have wildly different surroundings, so a single average is wrong for any one site.
- Measurement shows  $PL(d)$  at a given  $d$  is **random**, and **log-normally** distributed about the average.
- Add a zero-mean Gaussian random variable  $X_\sigma$  in dB, standard deviation  $\sigma$  dB:

$$PL(d) = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

$$P_r(d) [\text{dBm}] = P_t [\text{dBm}] - PL(d)$$

- Called **shadowing**: the mobile is shadowed by clutter (buildings, hills, foliage).
- $n$  and  $\sigma$  are found from measured data by **linear regression**, minimising mean square error.
- Small  $\sigma$  = the model fits that environment well.**

### Link budget design

- A link budget adds every gain and subtracts every loss between transmitter output and receiver input, then compares against the receiver **sensitivity**.
- Generic form, all terms in dB:

$$P_r = P_t + G_t + G_r - L_{\text{cable,t}} - L_{\text{cable,r}} - L_{\text{misc}} - PL(d)$$

- Link closes if  $P_r \geq \text{sensitivity} + \text{link margin}$ .
- Two** budgets are needed, forward and reverse, and the **weaker one sets the cell radius**.

### How to run the exam version

- Compute **maximum allowable path loss** on each link:

$$PL_{\text{max}} = P_t + G_t + G_r - L_{\text{cables}} - \text{sensitivity} - \text{margin}$$

- Put  $PL_{\text{max}}$  into the propagation model given (usually **Okumura**) and solve for  $d$ .
- Take the **smaller** of the forward and reverse distances.
- Compare against the range asked for and state whether the link is feasible.

### Two ways to build a propagation model

- Empirical**: collect measurements, fit curves. Eg Okumura, Hata.
  - Accurate in the environment measured, no guarantee elsewhere.
- Analytical**: model the physics, derive equations. Eg free space, two-ray.
  - Purely analytical models are inaccurate because so many assumptions are needed to keep the derivation tractable.
- Practical models are hybrids**: an analytical core with measured correction factors.

### 3.5 Outdoor propagation models

Explain any two outdoor propagation models. State the conditions for Okumura and Hata 3+3 5 2+2  
4

**TOP 7/22** 71 Ma, 73 Ma, 75 Bh, 79 Ch, 80 Ch, 81 Bh, 80 Bh · the numerical version appears in 13 of 22 papers → Ch 3 companion

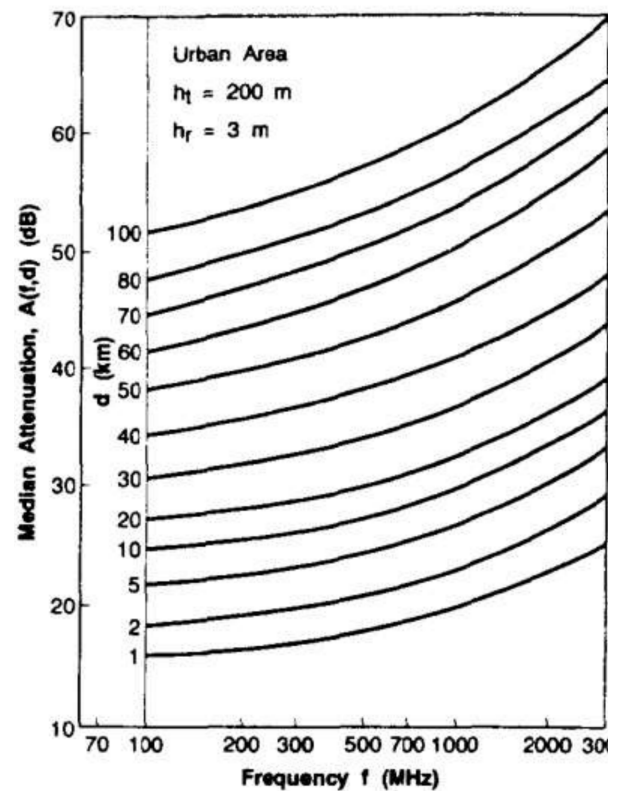
Outdoor transmission runs over irregular terrain, so the **terrain profile**, trees, buildings and hills must all be accounted for. Purely analytical models cannot do this, so all the outdoor models below are **empirical**.

#### Okumura model 4

- **Wholly empirical.** Built from extensive drive-test measurements in and around Tokyo, 1968.
- **The single most widely used model for signal prediction in urban areas.**
- Okumura published a set of **curves** of median attenuation relative to free space,  $A_{mu}(f, d)$ , for an urban area over quasi-smooth terrain, plus correction curves.
- **Validity conditions (the 2-mark answer)**
  - Frequency **150 MHz to 1920 MHz** (extrapolated to 3 GHz).
  - Distance **1 km to 100 km**.
  - BS antenna height **30 m to 1000 m**.
  - Quasi-smooth terrain, urban reference.
- **The formula**

$$L_{50}(\text{dB}) = L_F + A_{mu}(f, d) - G(h_{te}) - G(h_{re}) - G_{\text{AREA}}$$

- $L_{50}$  = median (50th percentile) path loss,
- $L_F$  = free space path loss,
- $A_{mu}$  = median attenuation relative to free space, read off the curve,
- $G(h_{te}), G(h_{re})$  = antenna height gain factors,
- $G_{\text{AREA}}$  = gain from the environment type.



$A_{mu}(f, d)$  over a quasi-smooth urban area,  $h_{te} = 200$  m,  
 $h_{re} = 3$  m.

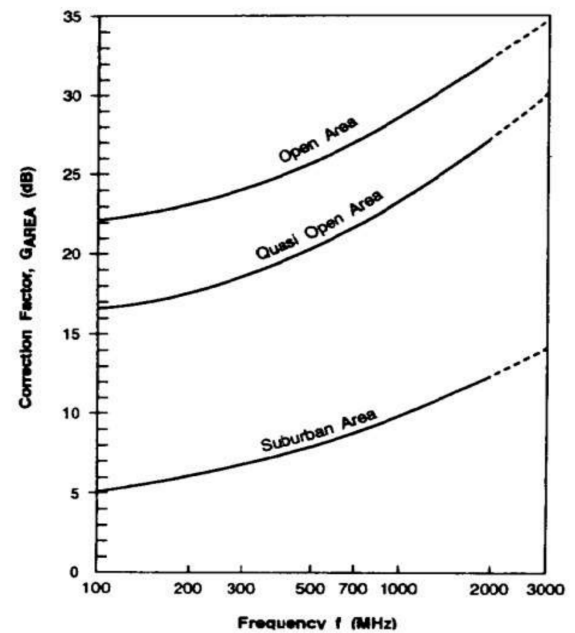
### – Height gain factors:

$$G(h_{te}) = 20 \log \frac{h_{te}}{200}, \quad 30 \text{ m} < h_{te} < 1000 \text{ m}$$

$$G(h_{re}) = 10 \log \frac{h_{re}}{3}, \quad h_{re} \leq 3 \text{ m}$$

$$G(h_{re}) = 20 \log \frac{h_{re}}{3}, \quad 3 \text{ m} < h_{re} < 10 \text{ m}$$

- (+) simple, and **the most accurate model available for mature cellular and land-mobile systems in cluttered urban areas.**
- (–) slow response to rapid terrain change → good in cities, poorer in rural areas. Deviation from measurement is 10 to 14 dB.
- (–) needs the curves, so it cannot be evaluated from formulas alone.



$G_{AREA}$  for suburban, quasi-open and open areas.

### Hata model 4 6 7

- **Hata is Okumura's curves turned into formulas**, so it can be worked by hand.
- Same measured data, so the same physical assumptions.
- **Validity conditions**
  - Frequency **150 MHz to 1500 MHz**.
  - Distance **1 km to 20 km**.
  - BS antenna **30 m to 200 m**, MS antenna **1 m to 10 m**.
  - **No path-specific correction** is possible, unlike Okumura.
- **Urban median path loss:**

$$L_{50}(\text{urban}) = 69.55 + 26.16 \log f_c - 13.82 \log h_{te} - a(h_{re}) + (44.9 - 6.55 \log h_{te}) \log d$$

with  $f_c$  in MHz,  $h_{te}$ ,  $h_{re}$  in m,  $d$  in km.

- **Mobile antenna correction  $a(h_{re})$ :**
  - small / medium city:  $(1.1 \log f_c - 0.7)h_{re} - (1.56 \log f_c - 0.8)$  dB
  - large city,  $f_c \leq 300$  MHz:  $8.29(\log 1.54h_{re})^2 - 1.1$  dB
  - large city,  $f_c > 300$  MHz:  $3.2(\log 11.75h_{re})^2 - 4.97$  dB
- **Suburban:**  $L_{50}(\text{urban}) - 2[\log(f_c/28)]^2 - 5.4$
- **Open rural:**  $L_{50}(\text{urban}) - 4.78(\log f_c)^2 + 18.33 \log f_c - 40.98$
- Suburban and rural terms are just **reductions in fixed loss** for less cluttered environments.

### Okumura Vs Hata (81 Bh asks exactly this, 4)

- **Form:** Okumura is graphical, Hata is closed-form.
- **Frequency:** Okumura to 1920 MHz, Hata to 1500 MHz.
- **Distance:** Okumura 1 to 100 km, Hata 1 to 20 km.
- **BS height:** Okumura 30 to 1000 m, Hata 30 to 200 m.
- **Path-specific corrections:** Okumura ✓, Hata ×
- **Accuracy:** within **1 dB of Okumura for  $d > 1$  km**, so Hata is preferred for large-cell work. Okumura is better for small cells and irregular terrain.
- **Extension:** COST-231 extends Hata to **2 GHz** by changing the constants, for PCS bands. verified/added

### Longley–Rice (ITS irregular terrain model)

- Point-to-point, **40 MHz to 100 GHz**, over irregular terrain.
- Runs as a **computer program**. Inputs: frequency, path length, polarisation, antenna heights, surface refractivity, ground conductivity and permittivity, climate.
- Two modes:
  - **point-to-point**, using a detailed terrain profile,
  - **area**, using empirical estimates where no profile exists.
- (–) **ignores multipath, buildings and foliage**, so it is a terrain model, not an urban model.

**Walfisch–Bertoni** syllabus, no PYQ yet

- Models **rooftop diffraction** explicitly, so it fits dense urban areas where the BS antenna is above the rooftops.
- Path loss is the product of three factors:

$$S = P_0 Q^2 P_1$$

- $P_0$  = free space loss between isotropic antennas,
- $Q^2$  = reduction from the **row of buildings** shadowing the street,
- $P_1$  = loss from the **final rooftop-to-street** diffraction.
- Predicts an overall  $\propto d^{-3.8}$  roll-off, matching urban measurement.
- Adopted as part of **COST-231**.

**Ericsson multiple breakpoint model** (80 Bh asks by name, 2)

- Built from measurements in a multi-floor office building.
- Uses a **uniform distribution** to generate path loss between a minimum and maximum bound at each distance.
- **4 breakpoints**, each carrying an upper and lower bound on path loss.
- Assumes **30 dB attenuation** at  $d_0 = 1$  m, accurate at 900 MHz w/ unity-gain antennas.
- Slope changes from **20 dB/decade** at the first breakpoint to **60 dB/decade** in the final section.
- **Its value is that it gives a deterministic limit on the range of path loss**, not a single number.

**3.6 Indoor propagation models****Explain indoor propagation models (any two). What factors influence them?** 8 5

PIN 2/22 70 Bh, 82 Bh

**How indoor differs from outdoor (the “factors” answer)**

- Distances covered are **much smaller**.
- Environment varies **far more** over a much smaller Tx–Rx separation.
- Propagation is governed by the **same three mechanisms**, but conditions are far more variable.
- **Factors that influence it:**
  - **Layout** of the building, room sizes, corridors.
  - **Construction materials:** concrete, metal, plaster, glass.
  - **Building type:** office, residential, factory.
  - Whether doors are **open or closed**.
  - Where the antenna is mounted: **desk height Vs ceiling**.
  - Number of **floors** between Tx and Rx.
  - Number of **windows** and whether they are tinted (metallised tint is a strong reflector).
  - Movement of **people and furniture**.
- **Path loss exponents indoors range from 1.6 (LOS corridor, better than free space, because of waveguiding) to 6 (obstructed).**

**Model 1: log-distance path loss (same form as outdoor)**

$$PL(d) [\text{dB}] = \overline{PL}(d_0) + 10n \log\left(\frac{d}{d_0}\right) + X_\sigma$$

- $n$  and  $\sigma$  are tabulated **per building type**.
- Smaller  $\sigma \rightarrow$  better accuracy.
- $d_0$  is typically **1 m** indoors.

**Model 2: attenuation factor model**

$$PL(d) [\text{dB}] = \overline{PL}(d_0) + 10 n_{SF} \log\left(\frac{d}{d_0}\right) + \text{FAF} + \sum \text{PAF}$$

- $n_{SF}$  = path loss exponent measured on the **same floor**.
- **FAF** = floor attenuation factor, added once per floor penetrated.
- **PAF** = partition attenuation factor, summed over the partitions crossed.
- **Standard deviation drops to about 4 dB**, against 13 dB for the plain log-distance model.

### Partition losses

- **Hard partitions:** structural walls of a room. Fixed.
- **Soft partitions:** movable dividers that do not reach the ceiling.
- **Between floors:** FAF is **not linear in floor count**. The first floor costs the most (about 15 dB), each further floor adds only 6 to 10 dB, then it saturates.
- Reason: signal also travels **out of and back into** the building via windows, so the direct path stops dominating.

| Material Type             | Loss (dB) | Frequency |
|---------------------------|-----------|-----------|
| All metal partition       | 26 dB     | 815 MHz   |
| Concrete block wall       | 13 dB     | 1300 MHz  |
| Empty Cardboard boxes     | 3-6 dB    | 1300 MHz  |
| Dry Plywood (0.75 inches) | 1 dB      | 9.6 GHz   |
| Dry Plywood (0.7 inches)  | 4 dB      | 28.2 GHz  |

Measured same-floor partition losses.

### 3.7 Small-scale fading and multipath

What is small-scale fading? Give its types and the factors that influence it 2+4+4 4+4 3+6 1+2+5 2+2

**TOP 6/22** 72 Ma, 73 Ma, 75 Bh, 78 Ch, 80 Ba, 81 Ba · the most repeated question in the chapter

#### Definition

- **Small-scale fading** = rapid fluctuation of the amplitude, phase and multipath delay of a radio signal over a **short travel distance or short time interval**.
- Caused by **interference between two or more copies** of the transmitted signal that arrive at slightly different times.
- The copies (multipath waves) add **vectorially** with random phases, so the sum behaves like noise.
- **Received power can change by 30 to 40 dB when the receiver moves a fraction of a wavelength.**
- Large-scale path loss can be ignored over such a short distance.

#### The three effects it produces

- **Rapid changes in signal strength** over a small travel distance or time interval.
- **Random frequency modulation** due to different **Doppler shifts** on different multipath components.
- **Time dispersion (echoes)** caused by multipath propagation delays → **ISI**.

#### Time dispersion Vs frequency dispersion

- **Time dispersion:** received signal **lasts longer** than the transmitted one, because paths have different delays. Measured by **delay spread**.
- **Frequency dispersion:** received signal is **wider in frequency** than the transmitted one, because paths have different Doppler shifts. Measured by **Doppler spread**.
- Both distort the received signal, and they are **independent** of each other. **That is why fading is classified twice over, once by each.**

#### Impulse response model of the multipath channel

syllabus, no PYQ yet

- The channel is treated as a **linear filter with a time-varying impulse response**.
- Feed it a unit impulse  $\delta(t)$  and the output is

$$h(t) = \sum_{n=1}^N A_n \exp(-j\phi_n) \delta(t - \tau_n)$$

- $A_n, \tau_n, \phi_n$  = amplitude, arrival delay and phase of the  $n$ -th path.
- Time variation comes from motion, so it is really  $h(t, \tau)$ :  $\tau$  indexes the multipath delay,  $t$  indexes the mobile's position.
- **This one model generates everything else in the chapter:** the PDP is  $|h(t, \tau)|^2$ , and the delay-spread parameters are its moments.

**Factors influencing small-scale fading (learn all 4)**

- **Multipath propagation.**
  - Random amplitude and phase of each component → the sum fluctuates.
  - Also **lengthens** the time the baseband signal takes to arrive → signal smearing → ISI.
- **Speed of the mobile.**
  - Relative motion causes a **Doppler shift**  $f_d$  on each component.
  - Different components arrive from different angles → different shifts → random FM.
- **Speed of surrounding objects.**
  - If nearby objects move **faster than the mobile**, they dominate the fading.
  - If they move slower, ignore them.
- **Transmission bandwidth of the signal.**
  - If signal bandwidth  $>$  channel bandwidth, the received signal is **distorted**, but its strength does not fade much over a local area.
  - If signal bandwidth  $<$  channel bandwidth, amplitude **changes rapidly** but the signal is not distorted in time.
  - **A channel gives you distortion or fading, typically one or the other.**

**Multipath measurement techniques** syllabus, no PYQ yet

- **Direct RF pulse sounding:** transmit a narrow pulse, display the received power against delay on a scope. Simple, but a single pulse can be lost in a fade.
- **Spread spectrum sliding correlator:** transmit a PN sequence, correlate at the receiver w/ a slightly slower clock so the correlation peak slides through the delays. Good dynamic range, rejects interference.
- **Swept frequency (frequency domain) sounding:** sweep a network analyser across the band, measure  $S_{21}$ , then inverse Fourier transform to get  $h(\tau)$ . Needs a cabled link, so it suits indoor work only.

**Define Doppler shift and Doppler spread. Derive the expression for Doppler shift** 2+4 1+2+5 2

**HOT 4/22** 78 Ch, 81 Ba, 80 Ch, 82 Bh · 82 Bh words it as "significance of frequency dispersion"

**Derivation (reproduce these 5 lines exactly)**

- Mobile moves at constant velocity  $v$  along a segment of length  $d$  from  $X$  to  $Y$ , receiving from a **distant** source  $S$ .
- Source is far away → the arrival angle  $\theta$  is the same at  $X$  and  $Y$ .
- **Path length difference**

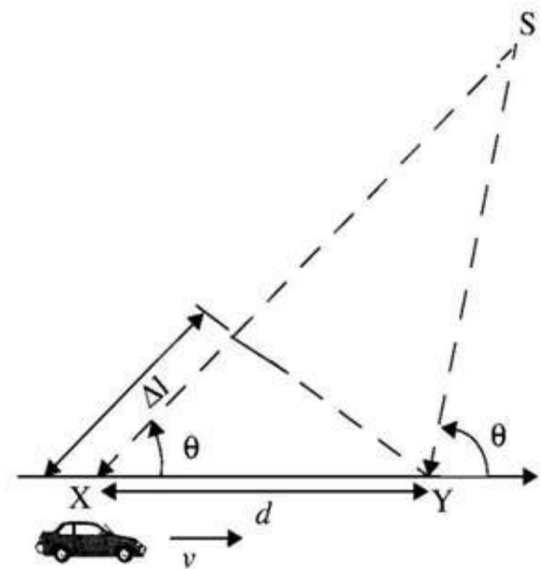
$$\Delta l = d \cos \theta = v \Delta t \cos \theta$$

- **Phase change** in the received signal

$$\Delta \phi = \frac{2\pi \Delta l}{\lambda} = \frac{2\pi v \Delta t}{\lambda} \cos \theta$$

- **Apparent frequency change** is the rate of change of phase:

$$f_d = \frac{1}{2\pi} \cdot \frac{\Delta \phi}{\Delta t} \Rightarrow \boxed{f_d = \frac{v}{\lambda} \cos \theta}$$



**Illustration of Doppler effect.**

Mobile moving from  $X$  to  $Y$  at velocity  $v$ ; source  $S$  is far away, so  $\theta$  is unchanged.

### Reading the result

- $\theta = 0^\circ$ , moving **towards** the source:  $f_d = +v/\lambda$ , maximum **positive** shift. Received frequency  $f_c + f_d$ .
- $\theta = 180^\circ$ , moving **away**:  $f_d = -v/\lambda$ , maximum negative shift.
- $\theta = 90^\circ$ , moving **perpendicular**:  $f_d = 0$ , no shift at all.
- **Maximum Doppler shift**

$$f_m = \frac{v}{\lambda} = \frac{vf_c}{c}$$

### Doppler spread $B_D$

- Measure of the **spectral broadening** caused by the time rate of change of the channel.
- Defined as the **maximum Doppler shift**:  
 $B_D = f_m = v/\lambda$ .
- A pure tone at  $f_c$  comes out spread over  $f_c \pm f_m$ .
- Characterises the channel's **frequency dispersiveness**.

### Explain the parameters of the mobile multipath channel 7 4 2+4

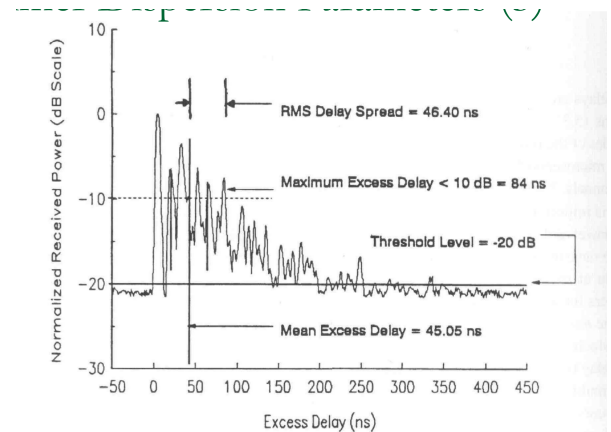
**HOT 3/22** 71 Ma, 80 Bh, 70 Ma · 80 Bh asks which parameters are used to **classify the types of fading** · **numericals on this appear in 5 papers** → Ch 3 companion

### Power delay profile (PDP)

- **PDP** = received power against **excess delay**, ie the intensity of the signal arriving through the multipath channel as a function of delay.
- Obtained by **averaging many impulse response measurements** over a local area.
- Every time-dispersion parameter below is a **moment of the PDP**.

### Significance of frequency dispersion (82 Bh)

- It makes the channel **time-varying**, so the receiver must keep re-estimating it.
- If  $B_D$  approaches the signal bandwidth, the channel changes **within one symbol** → **fast fading, and an irreducible BER floor that more transmit power cannot cure**.
- It sets how often an **adaptive equalizer must retrain** and how fast a **carrier recovery loop** must track.
- Different multipath components have different  $f_d$ , so the composite signal suffers **random FM**, heard as noise on an analog system.



A measured PDP w/ its mean excess delay, rms delay spread and 10 dB maximum excess delay marked. Note the -20 dB noise threshold.

### Time dispersion parameters

- **First arrival delay**  $\tau_A$ : delay of the first signal to reach the receiver, set by the shortest path. Used as the **reference**, so every other delay is an excess delay.
- **Mean excess delay** = first moment of the PDP:

$$\bar{\tau} = \frac{\sum_k a_k^2 \tau_k}{\sum_k a_k^2} = \frac{\sum_k P(\tau_k) \tau_k}{\sum_k P(\tau_k)}$$

- **rms delay spread** = square root of the second central moment:

$$\sigma_\tau = \sqrt{\overline{\tau^2} - (\bar{\tau})^2}, \quad \overline{\tau^2} = \frac{\sum_k a_k^2 \tau_k^2}{\sum_k a_k^2}$$

- **Maximum excess delay (X dB)** =  $\tau_X - \tau_0$ , the delay over which multipath energy stays within X dB of the strongest component.
- $\sigma_\tau$  **is the one that matters in practice**, because it, not  $\tau_{\max}$ , predicts ISI.
- These values **depend on the noise threshold** chosen. Set the threshold too low and noise is counted as multipath, inflating every parameter.

### Coherence bandwidth $B_c$

- **Definition**: the range of frequencies over which the channel passes all spectral components with **approximately equal gain and linear phase**, ie over which the channel looks **flat**.
- It is the **frequency-domain dual** of the rms delay spread. There is no exact relationship, only rules of thumb:

$$B_c \approx \frac{1}{50 \sigma_\tau} \quad (0.9 \text{ correlation, "90 \%"})$$

$$B_c \approx \frac{1}{5 \sigma_\tau} \quad (0.5 \text{ correlation, "50 \%"})$$

- Two sinusoids separated by **more** than  $B_c$  are affected **differently** by the channel.
- **Significance**: compare  $B_c$  against the signal bandwidth to decide whether the channel is **flat or frequency selective**. That single comparison decides whether an equalizer is needed.

### Coherence time $T_c$

- **Definition**: the time over which the channel impulse response is essentially **time-invariant**.
- Dual of the Doppler spread:

$$T_c \approx \frac{1}{f_m}, \quad T_c \approx \sqrt{\frac{9}{16\pi f_m^2}} = \frac{0.423}{f_m}$$

The second (geometric mean) form is the one normally used.

- Two signals arriving more than  $T_c$  apart are affected **differently**.
- **Large**  $T_c \rightarrow$  channel changes slowly.
- **Significance**: compare  $T_c$  against the symbol period to decide whether the channel is **slow or fast fading**. Also sets how often the equalizer must retrain.

## Explain the types of small-scale fading 4+4 1+3 2+7 2+4

**TOP 6/22** 70 Ma, 74 Ma, 79 Ch, 81 Ba, 80 Ch, 80 Bh · fading is classified twice over: once by delay spread, once by Doppler spread. Always give both tables.

### A. By multipath time delay spread

|                        | Flat fading                         | Frequency selective  |
|------------------------|-------------------------------------|----------------------|
| Bandwidth              | $B_S \ll B_C$                       | $B_S > B_C$          |
| Delay spread           | $\sigma_\tau \ll T_S$               | $\sigma_\tau > T_S$  |
| Spectrum of the signal | <b>preserved</b>                    | <b>not preserved</b> |
| Channel acts as        | flat gain, linear phase             | a filter             |
| ISI                    | none                                | <b>severe</b>        |
| Cure                   | raise transmit power, use diversity | <b>equalizer</b>     |

- **Flat fading** is the most common type.
- Signal bandwidth is narrow, so the whole signal rides up and down together.
- Causes **deep fades**, so flat-fading channels are sometimes called **amplitude varying** or **narrowband** channels.
- **Frequency selective fading** occurs when the delayed copies overlap **into the next symbol**.
- **Frequency selective fading is harder to model**, because the channel response depends on the exact multipath structure.

### B. By Doppler spread

|                 | Fast fading                                           | Slow fading                                           |
|-----------------|-------------------------------------------------------|-------------------------------------------------------|
| Doppler spread  | high, $B_S < B_D$                                     | low, $B_S \gg B_D$                                    |
| Coherence time  | $T_C < T_S$                                           | $T_C \gg T_S$                                         |
| Channel changes | <b>within</b> a symbol                                | over <b>many</b> symbols                              |
| Relative rates  | channel varies <b>faster</b> than the baseband signal | channel varies <b>slower</b> than the baseband signal |

- Fast / slow says **nothing** about flat / selective. **The four combinations are all possible**: flat slow, flat fast, frequency selective slow, frequency selective fast.
- **Fast fading** arises from motion, or from motion of the surroundings, not from distance.
- A fast fading channel produces an **irreducible BER floor**: extra transmit power does not help, because the channel has already changed by the time the symbol ends.
- In practice fast fading only matters for **very low data rates**.

## 3.8 Rayleigh and Ricean fading

## Rayleigh fading channel Vs Ricean fading channel, with appropriate expressions 5 2 2+4

**HOT 5/22** 70 Ma, 71 Ma, 72 Ash, 73 Bh, 80 Ch

### What these distributions describe

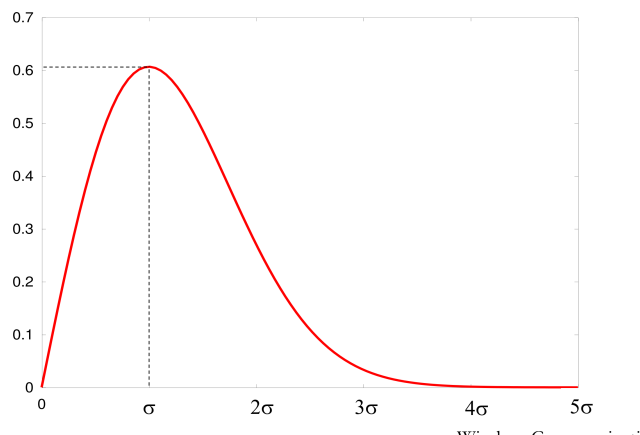
- Not *when* the signal fades, but **how the received envelope is statistically distributed**.
- Both apply to **flat fading** channels.

### Rayleigh

- Applies when **all multipath components have roughly equal amplitude** and there is **no dominant LOS component**.
- Physically: mobile deep in an urban area, far from the BS, no line of sight.
- The in-phase and quadrature sums are each Gaussian, so the **envelope** is Rayleigh:

$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) & 0 \leq r \leq \infty \\ 0 & r < 0 \end{cases}$$

- $\sigma$  = rms value of the received voltage **before** envelope detection.
- $\sigma^2$  = time-average power before envelope detection.
- **Phase** is uniformly distributed over 0 to  $2\pi$ .
- Key values:
  - mean  $r_{\text{mean}} = \sigma \sqrt{\pi/2} = 1.2533 \sigma$
  - median  $r_{\text{median}} = 1.177 \sigma$
  - variance  $\sigma_r^2 = 0.4292 \sigma^2$
- **Mean and median differ by only 0.55 dB**, which is why the median is used in practice.



### Ricean

- Applies when there is a **stationary, non-fading, dominant** component, normally a LOS path, on top of the random multipath.
- Physically: microcell, indoor pico-cell, or a mobile with clear sight of the BS.
- The random components add onto a fixed phasor, so:

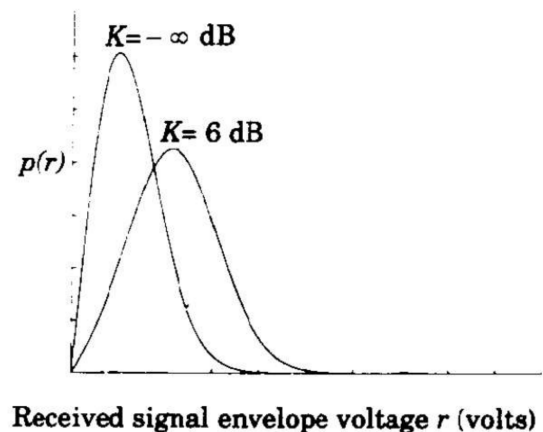
$$p(r) = \begin{cases} \frac{r}{\sigma^2} \exp\left(-\frac{r^2 + A^2}{2\sigma^2}\right) I_0\left(\frac{Ar}{\sigma^2}\right) & A \geq 0, r \geq 0 \\ 0 & r < 0 \end{cases}$$

- $A$  = peak amplitude of the **dominant** signal.
- $I_0(\cdot)$  = modified Bessel function of the first kind, zeroth order.
- **Ricean  $K$ -factor**

$$K = \frac{A^2}{2\sigma^2}, \quad K [\text{dB}] = 10 \log \frac{A^2}{2\sigma^2}$$

= ratio of **deterministic power** to **multipath power**.

- $A \rightarrow 0$ , **ie  $K \rightarrow -\infty$  dB: Ricean degenerates into Rayleigh**.
- $K \gg 1$ : the distribution tends to **Gaussian** about the mean, ie almost no fading.



$K = -\infty$  dB is the Rayleigh case;  $K = 6$  dB shows a dominant component pulling the distribution towards Gaussian.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short    | Means                                       | Short    | Means                                      |
|----------|---------------------------------------------|----------|--------------------------------------------|
| BS       | base station                                | BTS      | base transceiver station                   |
| MS       | mobile station                              | MSC      | mobile switching centre                    |
| BSC      | base station controller                     | MSK      | minimum shift keying                       |
| CCI      | co-channel interference                     | ACI      | adjacent channel interference              |
| SIR, S/I | signal to interference ratio                | SNR, S/N | signal to noise ratio                      |
| GoS      | grade of service                            | MAHO     | mobile assisted handoff                    |
| FCA      | fixed channel assignment                    | DCA      | dynamic channel assignment                 |
| LOS      | line of sight                               | NLOS     | non line of sight                          |
| FDMA     | frequency division multiple access          | TDMA     | time division multiple access              |
| CDMA     | code division multiple access               | OFDM     | orthogonal frequency division multiplexing |
| FDD      | frequency division duplex                   | TDD      | time division duplex                       |
| GSM      | global system for mobile communications     | AMPS     | advanced mobile phone system               |
| PCS      | personal communication system               | DAS      | distributed antenna system                 |
| BCCH     | broadcast control channel                   | EIRP     | effective isotropic radiated power         |
| ERP      | effective radiated power                    | RF       | radio frequency                            |
| Tx, TX   | transmitter                                 | Rx, RX   | receiver                                   |
| ISI      | intersymbol interference                    | PDP      | power delay profile                        |
| DFE      | decision feedback equalizer                 | MLSE     | maximum likelihood sequence estimation     |
| LMS      | least mean square                           | RLS      | recursive least squares                    |
| MSE      | mean square error                           | MMSE     | minimum mean square error                  |
| ZF       | zero forcing                                | MRC      | maximal ratio combining                    |
| EGC      | equal gain combining                        | BER      | bit error rate                             |
| PCM      | pulse code modulation                       | DPCM     | differential PCM                           |
| ADPCM    | adaptive differential PCM                   | LPC      | linear predictive coder                    |
| RPE-LTP  | regular pulse excited, long term prediction | CELP     | code excited linear prediction             |
| RELTP    | residual excited linear prediction          | MPE      | multipulse excited                         |
| VSELP    | vector sum excited linear prediction        | ATC      | adaptive transform coding                  |
| SBC      | sub-band coding                             | DCT      | discrete cosine transform                  |
| FEC      | forward error correction                    | ARQ      | automatic repeat request                   |
| BCH      | Bose–Chaudhuri–Hocquenghem code             | RS       | Reed–Solomon code                          |
| TCM      | trellis coded modulation                    | PN       | pseudo-noise                               |
| attr(s)  | attribute(s)                                | no.      | number                                     |
| freq     | frequency                                   | info     | information                                |
| ops      | operations                                  | config   | configuration                              |
| rm       | remove                                      | vs, Vs   | versus                                     |
| w/, w/o  | with, without                               | Eg       | example / for example                      |
| +Fig     | draw a figure                               | (I)      | importance / need / significance           |
| →        | leads to / therefore                        | +        | and                                        |

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams. Monospace year tags are BEI (EX 715); the rest are BEX (EX 751).

Tier chips over 22 papers, 2070–2082 BS: **TOP /22** = topic asked in 6 or more papers · **HOT /22** = 3 to 5 papers · **PIN /22** = 1 or 2 papers but worth  $\geq 6$  marks. Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.